Project 2: Predicting Above-Median Graduation Rates

Binary Logistic Regression Analysis

DSC 531 | IPEDS Data

1. Overview

This analysis builds a binary logistic regression model to predict whether a postsecondary institution's six-year graduation rate for baccalaureate students (150% of normal time) falls above the median graduation rate across all qualifying institutions. The sample is restricted to institutions with an incoming cohort of at least 200 students, yielding a working dataset of 1,225 institutions after removing U.S. territories.

2. Median Cutoff

The median graduation rate was computed using PROC MEANS on all institutions meeting the cohort size criterion. The raw median was 0.5991402, which rounds to 0.599 (59.9%) when carried to three decimal places as required. The binary outcome variable above_median was coded 1 if an institution's graduation rate exceeded 0.599, and 0 otherwise.

The resulting binary split was well-balanced: 623 institutions (49.76%) fell at or below the cutoff and 629 institutions (50.24%) fell above it. After excluding U.S. territories (27 observations) and observations with missing predictor values, the final analytic sample used in model estimation was n = 1,029.

3. Specification Sheet

The table below documents all variables considered as potential predictors, their type, whether they were included in model selection runs, and the rationale for inclusion or exclusion.

Variable	Description	Type	Status	Notes
tuition2	In-state tuition (full-time undergrads)	Continuous	Included	Primary cost signal for institution type
fee2	In-state required fees (full-time undergrads)	Continuous	Included	Additional cost burden beyond tuition
avg_faculty_salary	Average faculty salary	Continuous	Included	Proxy for institutional investment in instruction

Variable	Description	Type	Status	Notes
cohort_total	Total incoming cohort size	Continuous	Included	Proxy for institutional scale
control	Control of institution (Public/Private NP/Private FP)	Categorical	Excluded	Dropped in backward selection (p=0.730)
hloffer_collapse	Highest level of offering (collapsed)	Categorical	Excluded	Dropped in backward selection (p=0.882)
locale_collapse	Locale type (collapsed)	Categorical	Excluded	Dropped in backward selection (p=0.231)
c21enprf_collapse	Undergraduate enrollment profile (collapsed)	Categorical	Excluded	Dropped in backward selection (p=0.105)
cbsatype	CBSA metro/micro type	Categorical	Excluded	Dropped in backward selection (p=0.614)
room	On-campus housing available (Y/N)	Categorical	Excluded	Dropped in backward selection (p=0.472)
board_collapse	Meal plan available (collapsed)	Categorical	Excluded	Borderline at p=0.10; excluded from final model
region	Census region (Northeast/Midwest/South/West)	Categorical	Excluded	Borderline at p=0.10; excluded from final model
tuition1	In-district tuition	Continuous	Not modeled	Correlated 0.9998 with tuition2; redundant
fee3	Out-of-state required fees	Continuous	Not modeled	Correlated 0.863 with fee2; redundant
roomcap	Total dormitory capacity	Continuous	Not modeled	Correlated 0.849 with cohort_total; redundant
fips	FIPS state code (52 categories)	Categorical	Not modeled	Too many categories; collapsed into region variable

Note: Variables marked 'Not modeled' were excluded prior to model selection due to high multicollinearity with retained predictors. Variables marked 'Excluded' were entered into model selection but removed during the selection process.

4. Model Selection

4.1 Methods and Criteria Explored

Model selection was conducted using PROC LOGISTIC with three direction methods and two significance-level criteria, yielding six total runs:

- Methods: Forward selection, Backward elimination, Stepwise selection
- Criteria: p-value threshold of 0.05 (slentry/slstay = 0.05) and p-value threshold of 0.10 (slentry/slstay = 0.10)

All six runs are summarized below:

Method	Criteria	Variables Selected	AIC	-2 Log L	c-statistic	n
Forward	p = 0.05	tuition2, cohort_total, avg_faculty_salary, fee2	769.194	759.194	0.914	1029
Backward	p = 0.05	tuition2, cohort_total, avg_faculty_salary, fee2	769.194	759.194	0.914	1029
Stepwise	p = 0.05	tuition2, cohort_total, avg_faculty_salary, fee2	769.194	759.194	0.914	1029
Forward	p = 0.10	tuition2, cohort_total, avg_faculty_salary, fee2, region, board_collapse	766.174	748.174	0.916	1029
Backward	p = 0.10	tuition2, cohort_total, avg_faculty_salary, fee2, region, board_collapse	766.174	748.174	0.916	1029
Stepwise	p = 0.10	tuition2, cohort_total, avg_faculty_salary, fee2, region, board_collapse	766.174	748.174	0.916	1029

4.2 Candidate Model Comparison

The six runs produced two distinct candidate models. The p = 0.05 runs (all three methods) converged on the same four-variable model containing tuition2, cohort_total, avg_faculty_salary, and fee2. The p = 0.10 runs added region and board_collapse to this core set, but neither variable was individually significant at the 0.05 level within that model (region: p = 0.070; board_collapse: p = 0.089).

The AIC improvement from adding region and board_collapse was minimal (769.194 vs. 766.174, a difference of only 3.0 points), and the c-statistic was essentially unchanged (0.914 vs. 0.916). Given these marginal gains and the fact that both additional variables fell short of conventional significance, the four-variable model was selected as the final model.

5. Final Model

5.1 Model Equation

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log[P(above_median=1) / P(above_median=0)] = -8.0792 + 0.1338(tuition2_k) +
          0.0251(fee2_100) + 0.0477(salary_k) + 0.0745(cohort_100)
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Where tuition2_k = in-state tuition in thousands of dollars, fee2_100 = in-state required fees per $100, salary_k = average faculty salary in thousands of dollars, and cohort_100 = incoming cohort size per 100 students.

5.2 Parameter Estimates and Odds Ratios

Parameter	Unit	Estimate	Std Error	Wald Chi-Sq	p-value	Odds Ratio
Intercept	—	-8.0792	0.6247	167.243	<.0001	—
tuition2_k	Per $1,000	0.1338	0.00898	222.012	<.0001	1.143
fee2_100	Per $100	0.0251	0.00664	14.253	0.0002	1.025
salary_k	Per $1,000	0.0477	0.00725	43.205	<.0001	1.049
cohort_100	Per 100 students	0.0745	0.0102	53.110	<.0001	1.077

Model fit: AIC = 769.194, -2 Log L = 759.194, c-statistic = 0.914. Likelihood ratio test: Chi-Square = 664.14 (4 df), p < .0001. n = 1,029.

5.3 Interpretation of Effects

All four predictors are continuous and positively associated with the odds of a graduation rate above the median. Interpretations are given in scaled units for practical relevance:

- Tuition (tuition2_k): Each $1,000 increase in in-state tuition is associated with a 14.3% increase in the odds of being above the median graduation rate (OR = 1.143, 95% CI: 1.123-1.164, $p < .0001$). This is the strongest predictor in the model. Higher tuition institutions tend to be more selective and better-resourced, which likely drives higher completion rates.
- Cohort size (cohort_100): Each additional 100 students in the incoming cohort is associated with a 7.7% increase in the odds of above-median graduation (OR = 1.077, 95% CI: 1.056-1.099, $p < .0001$). Larger institutions may benefit from greater economies of scale and more robust student support infrastructure.
- Faculty salary (salary_k): Each $1,000 increase in average faculty salary is associated with a 4.9% increase in the odds of above-median graduation (OR = 1.049, 95% CI: 1.034-1.064, $p < .0001$). Faculty salary serves as a proxy for institutional investment in instruction quality.
- Required fees (fee2_100): Each $100 increase in required in-state fees is associated with a 2.5% increase in the odds of above-median graduation (OR = 1.025, 95% CI: 1.012-1.039, $p = 0.0002$). Though the effect is modest per unit, fees can represent a meaningful additional cost signal at the higher end of the range.

5.4 Model Performance

The model demonstrates excellent discriminatory ability, correctly ordering 91.4% of all pairs of one above-median and one below-median institution (c-statistic = 0.914, Somers' D = 0.827). The model correctly classifies institutions well above what would be expected by chance (50%), indicating strong predictive utility. The global null hypothesis that all coefficients are zero is decisively rejected (Likelihood Ratio Chi-Square = 664.14, $p < .0001$).

6. Final Model Selection Rationale

The four-variable model was chosen over the six-variable alternative for the following reasons:

- Parsimony: The two additional variables in the $p = 0.10$ model (region and board_collapse) were not statistically significant at the conventional 0.05 threshold. Including non-significant predictors inflates model complexity without a meaningful increase in explanatory power.
- Negligible improvement in fit: The AIC difference between the two models was only 3.0 points (769.194 vs. 766.174), and the c-statistic improved by only 0.002 (0.914 vs. 0.916). Neither difference is practically meaningful.
- Consistency across methods: All three selection methods (forward, backward, stepwise) agreed on the same four-variable model at the 0.05 threshold, providing strong evidence that this set of predictors is stable and not an artifact of a particular selection strategy.
- Interpretability: The four retained predictors all have clear, substantively meaningful interpretations relating to institutional cost and scale. The final model tells a coherent story: higher-cost, larger, and better-paying institutions are more likely to graduate students at above-median rates.

Data source: IPEDS (Integrated Postsecondary Education Data System). Analysis conducted in SAS Viya.